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D

When the switch is open, at balance position, there is no current passing through the unknown emf cell.

$$E = \frac{0.70}{1.0} \times 2.0 = 1.4 \text{ V}$$

When switch is closed, at balance position, there is current flowing in the secondary circuit below.

$$E - Ir = \frac{0.50}{1.0} \times 2.0$$

$$1.4 - \left(\frac{1.4}{2.0 + r} \right) r = 1.0$$

$$r = 0.80 \, \Omega$$

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A

From $\nabla \times B = \frac{\mu_0 NI}{\dots}$